

WALMART SALES ANALYSIS

SQL Exploration & Power BI Dashboard

Dataset: 9,969 transactions | 100 branches | 98 cities

Date Range: January 2019 – December 2023

Tools: Python (data cleaning) | SQLite (analysis) | Power BI (visualization)

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1. Project Overview

This project performs an end-to-end sales analysis on a Walmart retail dataset. Raw transaction data was first cleaned and standardized using Python, then explored through a series of SQL queries in SQLite to answer key business questions. The findings were finally visualized in an interactive Power BI dashboard.

The goal is to demonstrate a complete data analytics workflow — from raw data to business insights — using industry-standard tools.

Tech Stack

Tool	Purpose
Python (pandas)	Data cleaning, null handling, type standardization
SQLite	SQL-based business analysis across 9 queries
Power BI	Interactive dashboard with KPI cards and charts

2. Dataset

The dataset contains 9,969 Walmart sales transactions across 100 branches in 98 cities in Texas, spanning January 2019 to December 2023. Each record captures invoice details, product category, payment method, quantity, pricing, and customer rating.

Key Metrics

\$1.21M Total Revenue	\$476K Total Profit	9,969 Transactions	23,483 Units Sold
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Dataset Columns

Column	Description
invoice_id	Unique transaction identifier
Branch	Store branch code (e.g. WALM003)
City	City where the branch is located
category	Product category (6 types)

Column	Description
unit_price	Price per unit (\$)
quantity	Number of units sold
date	Transaction date (DD/MM/YY)
time	Transaction time (HH:MM:SS)
payment_method	Ewallet, Cash, or Credit card
rating	Customer satisfaction rating (1–10)
profit_margin	Profit margin as a decimal
total	Total transaction value (\$)

3. Data Cleaning (Python)

The raw dataset was cleaned in Python using pandas before any SQL analysis was performed. The following steps were applied:

- Removed duplicate invoice records
- Handled missing values in rating and payment_method columns
- Standardized date formats to DD/MM/YY for consistent SQLite parsing
- Ensured numeric columns (unit_price, quantity, total, profit_margin) had correct data types
- Trimmed whitespace from string columns (Branch, City, category)
- Exported the cleaned dataset as cleaned_walmart_data.csv for use in SQLite and Power BI

4. SQL Analysis

Nine business questions were answered using SQL queries in SQLite. The queries range from basic aggregations to window functions and CTEs.

Queries Summary

#	Question	Technique Used
Q1	Transactions & quantity sold by payment method	GROUP BY, SUM
Q2	Highest-rated category per branch	Window function — RANK() PARTITION BY

#	Question	Technique Used
Q3	Busiest day of week per branch	CASE + strftime, RANK()
Q4	Total quantity sold per payment method	GROUP BY, SUM
Q5	Avg / min / max rating by city & category	GROUP BY, aggregate functions
Q6	Total profit per category	SUM with derived column
Q7	Most common payment method per branch	CTE + RANK() PARTITION BY
Q8	Sales volume by time of day (Morning / Afternoon / Evening)	CASE + strftime
Q9	Top 5 branches with highest revenue decline (2022→2023)	CTE + JOIN + calculated ratio

Queries Q2, Q3, and Q7 use window functions (RANK() OVER PARTITION BY) — a more advanced SQL technique for ranking within groups without collapsing rows. Q9 uses two CTEs to calculate year-over-year revenue and joins them to derive a decline ratio.

5. Key Findings

Payment Methods

- Credit card is the most used payment method with approximately 4,300 transactions, followed by Ewallet (3,900) and Cash (1,800).
- Credit card and Ewallet together account for over 80% of all transactions.

Product Categories

- Fashion accessories and Home and lifestyle are the top two profit-generating categories, each contributing ~\$192K in profit.
- Health and beauty is the lowest-profit category at ~\$19K, despite likely having a reasonable transaction volume.
- To maintain the high profit in Fashion and lifestyle categories, more variety and a higher supply can be provided to boost sales further.
- The lowest revenue generating category—Health and beauty needs to be worked on. This can be done by supplying higher quality, lower price items, or by offering sales to get people interested in them. The marketing team needs to shine more light on products in this category.

Sales by Time of Day

- Afternoon is the busiest period across all three displayed branches (WALM058, WALM009, WALM030).

- Morning consistently has the lowest transaction volume, suggesting opportunities for morning promotions.

Revenue Decline (2022 → 2023)

The five branches with the largest revenue declines are:

Branch	2022 Revenue	2023 Revenue	Decline Ratio
WALM047	\$2,581	\$1,069	58.58%
WALM045	\$1,731	\$647	62.62%
WALM098	\$2,446	\$1,030	57.89%
WALM033	\$2,099	\$931	55.65%
WALM081	\$1,723	\$850	50.67%

WALM045 shows the steepest decline at 62.62%, with revenue dropping from \$1,731 to just \$647. These branches are flagged as priority cases for investigation.

6. Power BI Dashboard

The Power BI dashboard consolidates the SQL findings into a single-page interactive report. It includes four KPI summary cards at the top and four visuals covering profit by category, payment method transactions, sales by time of day, and the revenue decline alert table.



Dashboard Components

Visual	Data Source	Insight
KPI Cards (x4)	Q1, Q6 aggregates	Top-line revenue, profit, transactions, quantity
Profit by Category (bar)	Q6	Fashion & Home dominate profit
Transactions by Payment (bar)	Q1	Credit card leads by significant margin
Sales by Time of Day (stacked bar)	Q8	Afternoon peak across all branches
Revenue Decline Table	Q9	Branches with biggest YoY drops flagged

7. Skills Demonstrated

Skill Area	Specifics
Data Cleaning	Python / pandas — deduplication, null handling, type casting, standardization
SQL — Intermediate	Aggregations, GROUP BY, ORDER BY, CASE statements, date parsing with strftime
SQL — Advanced	Window functions (RANK, PARTITION BY), CTEs, multi-CTE JOINS
Data Visualization	Power BI KPI cards, bar charts, stacked bars, conditional formatting tables
Business Thinking	Framed raw data into 9 actionable business questions with decision-relevant outputs
End-to-End Workflow	Moved data from raw CSV → cleaned dataset → SQL analysis → visual dashboard

8. How to Reproduce

The full project is available on GitHub. To run it locally:

- Clone the repository and open cleaned_walmart_data.csv in DB Browser for SQLite or any SQLite client
- Run the SQL queries from the queries.sql file in order — each query is labeled Q1 through Q9
- Open the Power BI .pbix file and refresh the data source to point to your local cleaned_walmart_data.csv
- All Python cleaning steps are in the data_cleaning.py notebook

End of Report